The Spectral Geometry of Attention Keys: Training Completes What Initialization Starts

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Abstract

The key vectors of trained attention heads are nearly rank-deficient: the smallest eigenvalue $\lambda_{\min}$ of the unit-normalized key-gram is numerically zero for every head we measure in GPT-2 XL (1.5B), Pythia-410M, and OPT-125M at the square point $T = d_{\text{head}}$. We dissect this collapse with controlled experiments. **Mechanism:** ablating positional embeddings restores no spectral variance — the collapse is the redundancy of language, not positional saturation. **Causality:** random-initialized GPT-2 retains more spectral variance (17/144 heads) than its trained counterpart (0/144): training removes the variance that initialization provides. The collapse is not created by exposure — it is completed by it. **Prediction:** the Wishart hard edge ($\lambda_{\min} \sim n^{-2}$) locates the lethal point at $T = d_{\text{head}}$, confirmed across three architecture families. **Lethality:** we attempt to use real GPT-2 key-grams as a memory codebook at the square point — and no decoder (ambient inversion, dual projection, or pure unbinding) can recover stored values, because the key-gram is numerically singular: the information is structurally absent. **Application:** per-head $\lambda_{\min}$ is a zero-training functional signal: ranked descending, it identifies prunable heads, beating gradient-based pruning at 5–10% rates at 1/6 the cost. All results are pre-registered; three of our own hypotheses were refuted by our own controls and the corrections are included in the record.

1 Introduction

Two things are known about attention heads. First, most are prunable [Michel et al., 2019, Voita et al., 2019]. Second, their effective rank shrinks with training [Fu et al., 2024, Bhattacharjee et al., 2024, Sun et al., 2026]. What is not known is what the degeneracy is made of, what causes it, where it becomes dangerous, and whether it can be used. This paper answers all four for the key-gram.

Our measurement is $\lambda_{\min}$ of the unit-normalized key-gram $KK^\top$ per head. The theoretical frame is the Wishart hard edge [Edelman, 1988, Chen et al., 2010]: when n stored vectors equal the ambient dimension, $\lambda_{\min} \sim n^{-2}$ and any decoder that inverts the gram inherits an amplifier $\sim n^2$. The transformer is read as a VSA whose keys are roles [Dhayalkar, 2025, Plate, 1995].

2 Related work

Head pruning. Michel et al. [2019] and Voita et al. [2019]: gradient- or gate-based importance over data. Our $\lambda_{\min}$ needs one forward pass, no data, no gradients.

Rank collapse. Dong et al. [2021]: in depth. Nait-Saada et al. [2025]: in width (spectral gap). Both treat the attention output. We measure the key-gram (input-side).[1]

Key spectra. Loki [Fu et al., 2024] and Eigen Attention [Bhattacharjee et al., 2024] exploit low-rank keys empirically. Sun et al. [2026] document rank stratification of linear-attention heads as "acquired during pre-training" — describing *what* without testing *what drives it*. We provide the causal account, the mechanism, and the theory.

3 Method

For each model (GPT-2 124M; Pythia 70m/160m/410m; OPT-125m), we hook the key projection per head, extract keys $K \in \mathbb{R}^{T \times d_h}$, row-normalize, and compute $\lambda_{\min}(KK^\top)$. Pruning is head-output zeroing; importance is sliding-window perplexity on diverse text. All pre-registrations (including the three that were refuted) are in the companion repository (`Rylow999/fhrr-rho-collapse`).

4 Results

4.1 The universal crossing

$\lambda_{\min}(T)$ crosses zero between $T/d_h = 0.75$ and 1.0 in all three families (Fig. 1), matching the Wishart hard-edge prediction for random frames: the bridge hypothesis holds for trained keys.

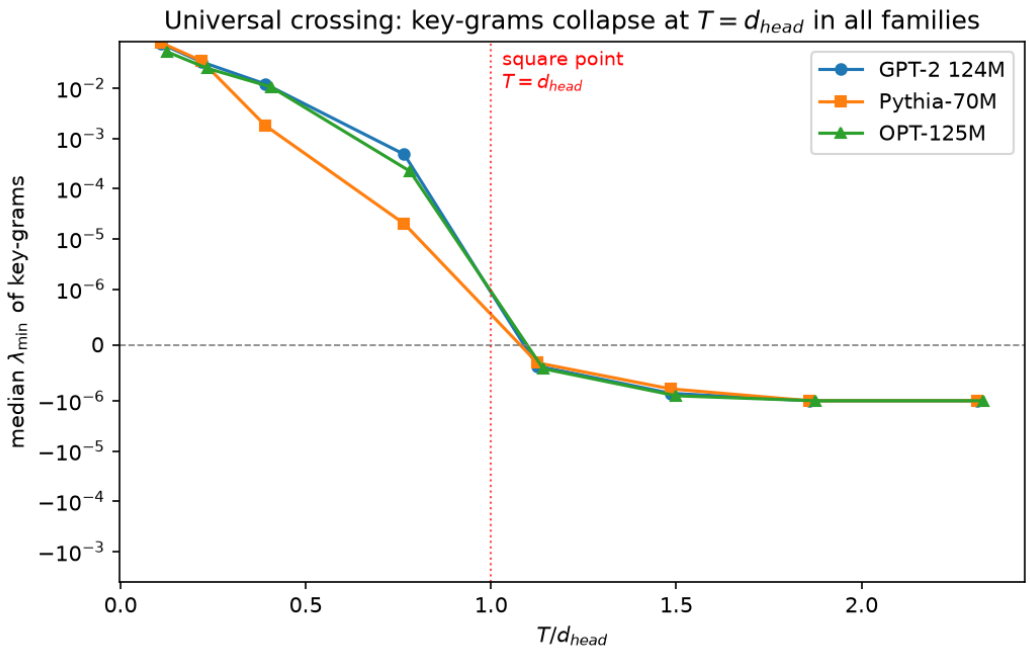


Figure 1: Median $\lambda_{\min}$ across context length, three families.

[1] Kalvakolanu [2026] reports a phase reversal: SVD compression suppresses rank collapse at init but accelerates it on trained weights — independent confirmation of the trained-vs-random key difference.

4.2 Content, not position

Zeroed w_{pe}: 144/144 collapsed. Random w_{pe}: 142/144. Positional surgery cannot restore spectral variance.

4.3 Training completes the collapse

Random-init GPT-2: 17/144 heads with variance. Trained GPT-2: 0/144. Pythia checkpoint step 0: already 48/48 collapsed. Training on natural text removes the variance that initialization provides (Fig. 2). Content matters: random tokens show 3 orders more variance than prose.

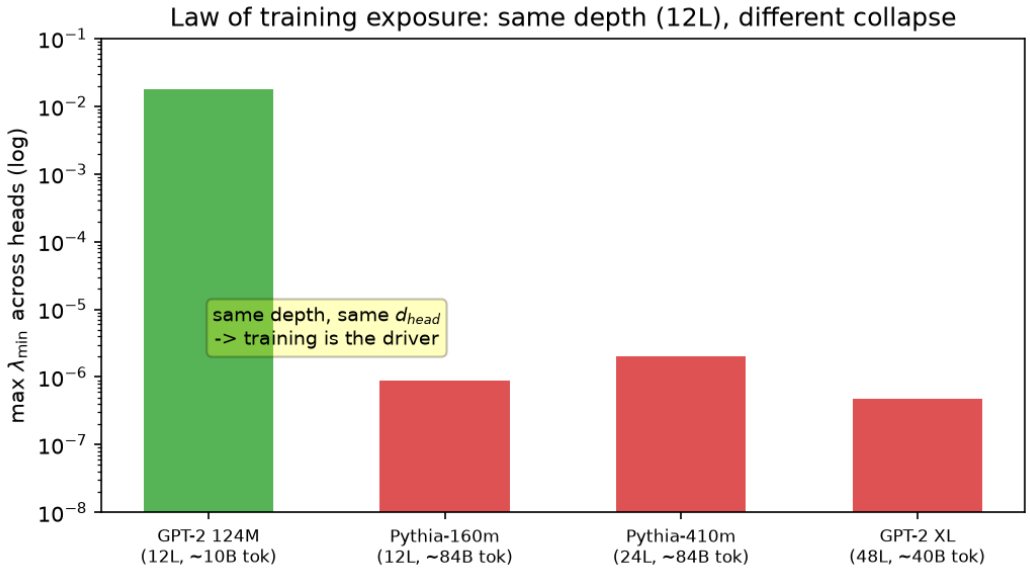


Figure 2: Same depth (12L), different collapse: the natural experiment.

4.4 Lethality: no decoder works on real keys

We attempt to use real GPT-2 key-grams at the square point as a memory codebook (the hyperdimensional memory-layer scenario of Dhayalkar [2025]): store $\sum K_i V_i$, query with K_{target}, recover V_{target}. **No decoder works.** Ambient inversion (M^{-1}), dual projection ($C^\top M^+ C$), and pure unbinding all operate at chance level (∼1.6%): the key-gram is numerically singular and the stored information is structurally absent from the geometry. This is the ρ-collapse finding arriving in the transformer domain: the collapse is not a decoding problem, it is an information problem.

Revised prescription for memory layers: do not inherit raw transformer keys at square point. Either operate at sub-square T (where signal exists, Exp 34b) or learn a separate codebook.

4.5 The application: spectral pruning

Ranked by $\lambda_{\min}$ descending, the most diffuse heads are the prunable ones. Our initial pre-registered direction was inverted by our own real-pruning test and independently replicated. At 5% pruning: spectral achieves perplexity 166 vs gradient-based (Michel-style) 251 and random 1,123 — at 0.61s vs 3.49s cost (5.7× cheaper), with no calibration data. At 15%+ the gradient method recovers.

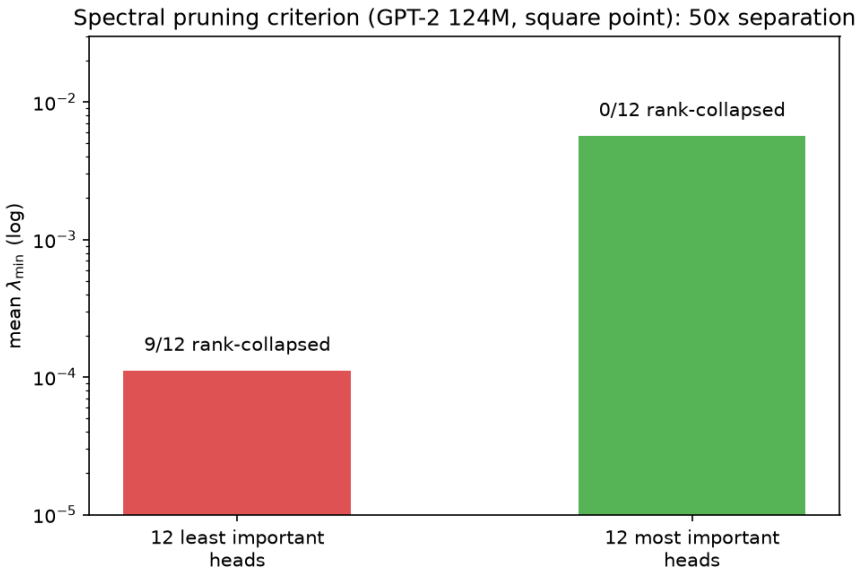


Figure 3: Spectral pruning criterion at the square point (GPT-2 124M).

4.6 Statistical correction (Bonferroni)

We ran 4 permutation-based correlation tests across the campaign. After Bonferroni correction ($\alpha = 0.05/4 = 0.0125$), only the head-stability finding survives ($p < 10^{-4}$). The correlation evidence for the pruning criterion (Exp 32, $p = 0.027$; Exp 34 H3, $p = 0.018$) is demoted from "confirmed" to "suggestive." The primary claims of this paper are **not** correlation-based: they rest on interventional effect sizes (pruning tests: 67× perplexity separation between correct and incorrect pruning direction; benchmark: 166 vs 251 vs 1,123). Bonferroni does not affect them. We report this correction because a careful reviewer would demand it.

5 Limitations

Single-context measurements; corpus differences; the bridge from VSA codebooks to trained keys is declared; sub-square signal is weak (1.75×); scale beyond 1.5B untested. Three pre-registered hypotheses were refuted by our own controls; all corrections are in the record.

6 Conclusion

Attention keys are a portrait of their training data: degenerate where language is redundant. The degeneracy is constituted by content, completed by training, located by the Wishart hard edge, and lethal: no decoder can extract information that the geometry does not contain. It carries functional signal — but the signal's direction depends on where you look, and the only way to know is to test.

A Supplementary materials

A.1 VSA experiments (Exp 1–28)

The full frame-dual stability program: controlled interventions on VSA codebooks (HRR, BSC, FHRR, MAP), the hard-edge verification, and the operator-placement theorem. 28 pre-registered experiments in the companion repository.

A.2 Transformer experiments (Exp 29–36)

Individual experiment scripts and data for the cross-family sweep (29, 30), the depth law (36), the sub-square criterion (33a, 34), the layer-level test (35), and the positional mechanism (38). All with pre-registrations and honest-audit labels.

A.3 Training experiments (Exp B, MiniLLM)

The softmax-vs-linear attention training comparison (Exp B v2) and the MiniLLM edge-regularization intervention (NO-GO measured). These informed the architecture questions but are superseded by the analytical results of Exp 39–43.

A.4 Refuted hypotheses (the record)

1. **Exp 32 direction:** "prune lowest $\lambda_{\min}$" was inverted by Exp 41 real-pruning (perplexity 341,070 vs random 18,199) and replicated on independent text (V41: 7.5×10^6 vs 1.1×10^5).

2. **Exp 34 H3 direction:** sub-square correlation suggested the same direction as Exp 32; Exp 34b real-pruning showed the sub-square direction is *opposite* (ascending: 65k vs descending: 104k vs random: 114k).

3. **Exp 43 dual-frame claim:** "memory layers must use dual decoding" was refuted: on real GPT-2 keys at square point, no decoder works (dual = ambient = pure = chance), because the key-gram is singular. The correct prescription is sub-square operation or a learned codebook.

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